

On MIR, Audio Descriptors and Classification

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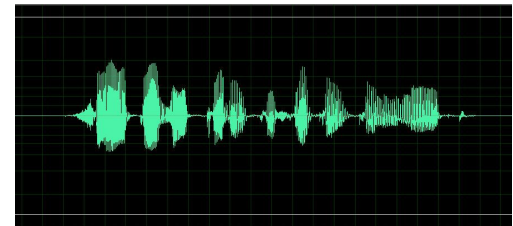
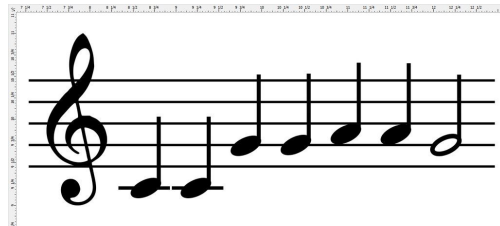
Music Information Retrieval

- Information Retrieval: extracting meaningful information from **data**.
- Music Information Retrieval: retrieving meaningful information from **music**

Schedl, M., Gomez, E., & Urbano, J. (2014). Music Information Retrieval: Recent Developments and Applications. Foundations and Trends in Information Retrieval (Vol. 8). <http://doi.org/10.1561/15000000045>

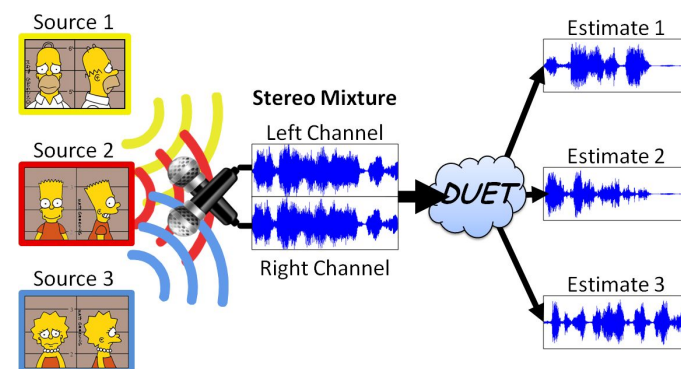
Music Information Retrieval

- Multidisciplinary field (psychology, computer science and engineering, musicology, **arts**)
- Retrieving meaningful information from music:
 - Symbolic formats (MIDI, scores, MusicXML)
 - Audio formats (WAV, MP3 etc.)
 - Live Performance, real-time input



Music Information Retrieval: Applications

- Recommender Systems
- Source Separation
- Genre Detection
- Mood Classification
- Transcription
 - Audio to MIDI
 - Chord Detection
 - Melody Extraction etc.
- **Musical Creativity**
- **Arts**



MIR in Music Creation

- MIR algorithms are part of many music making tools

- **Ableton**

- Onset detection
- Pitch detection
- Drum classification
- Harmony Extraction

- **Traktor**

- Bars & Beats Detection
- No more need for beat matching!



Common MIR Tasks

- Feature Extraction

- Extracting meaningful descriptors or features
- “A content descriptor is defined in the MPEG-7 standard as a distinctive characteristic of the data which signifies something to somebody”
- Scalar or vector
- Low-level or High-level

- Similarity Measurement

- Finding similar songs, sounds, artists etc.

- Classification

- Identifying which category something belongs (genre, instrument)

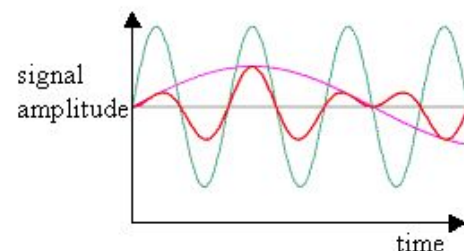
Feature Extraction - Taxonomy

- Metadata
 - filenames, tags, comments etc.
- Low-level descriptors
 - Signal processing derived (RMS, zero-crossing rate, etc).
- Mid-level
 - Derived features that apply musicological/perceptual knowledge (pitch, etc)
- High-level
 - Derived Features that are meaningful to humans
 - Mood, danceability, “mixability”, good/bad

Feature Extraction - Time vs. Frequency

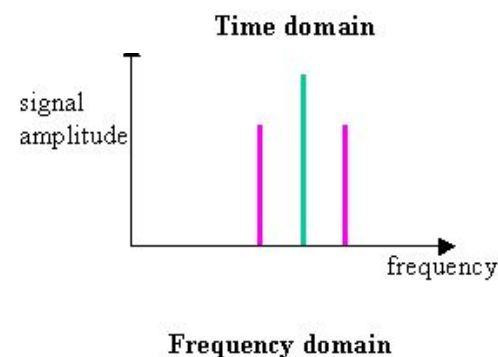
- Time Domain

- Energy
- ZCR (Zero Crossing Rate)
- Envelopes



- Frequency Domain

- Timbral descriptors (MFCCs)
- Harmonic descriptors
- Energy content in frequency bands

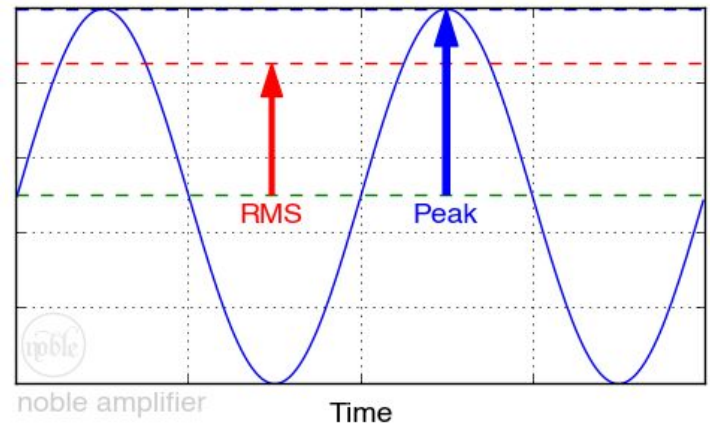


- Complex descriptors will often use a combination

Time-based descriptors - amplitude

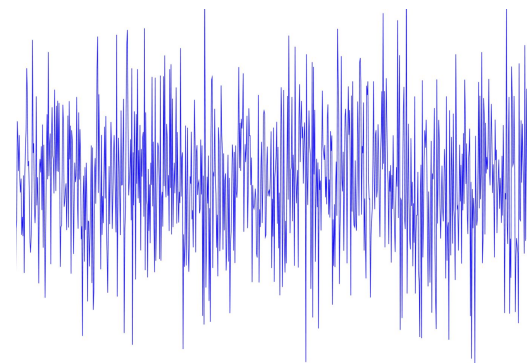
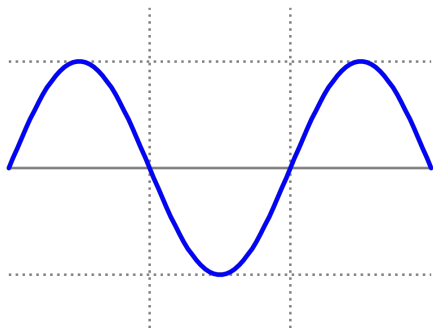
- Peak level is an “easy” descriptor value!
- But not a good descriptor, RMS is a much more accurate descriptor for how we perceive overall amplitude.

$$x_{\text{rms}} = \sqrt{\frac{1}{n} (x_1^2 + x_2^2 + \cdots + x_n^2)}.$$



Time-based descriptors - zero crossing rate

- Measure the number of times a signal crosses 0.
- Useful for telling how noisy a signal is
- Acts as a very naive pitch detector



$$zcr = \frac{1}{T-1} \sum_{t=1}^{T-1} \mathbb{1}_{\mathbb{R}_{<0}}(s_t s_{t-1})$$

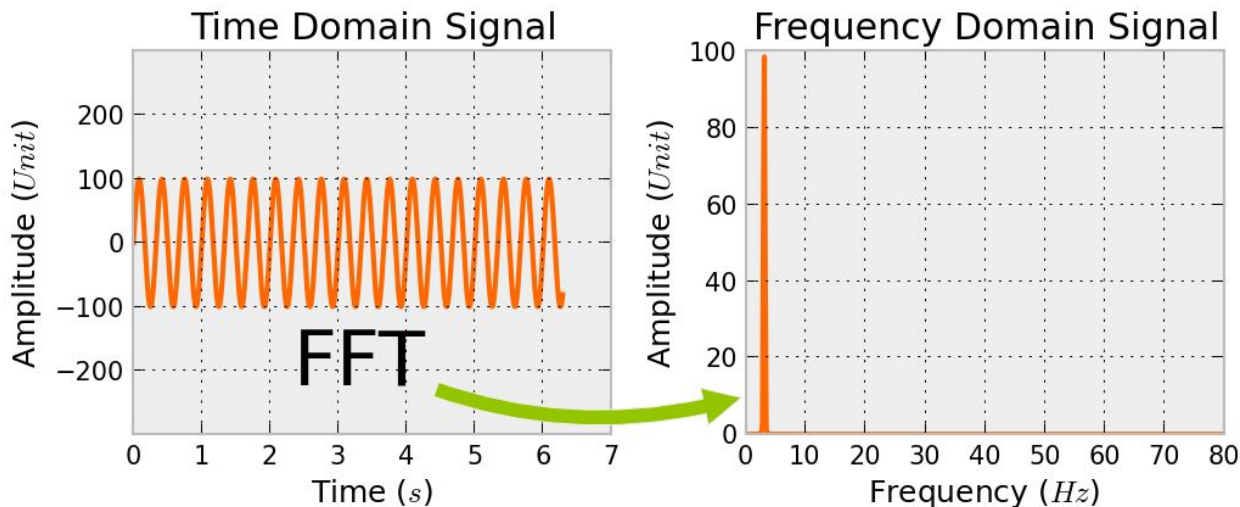
Time-based descriptors - global descriptors

- Analysis takes into account entire signal
- Log-Attack Time
 - Log of the time it takes to reach maximum intensity
- Temporal Centroid
 - Location of the centre of energy in the signal



Spectral descriptors

- Many descriptors are computed in the frequency domain.
- This is why the FFT and STFT are the focus of so much study in MIR signal processing.



Timbre: definitions

- Qualities which allow us to distinguish two different sound sources (e.g. instruments) playing the same pitch....
- Timbre is complex to describe not only for humans but also for computers.
- At the very least, **we can say that the notion of timbre is very connected to its spectral components and the temporal (envelope) profile of those components.**

Spectral descriptors

- **Brightness**
- **Centroid**
- **Flatness**
- **Flux** (change in energy per bin per frame)
- **Irregularity** (changes across neighbouring bins)
- **Kurtosis** (peakedness of spectrum)
- **Spectral rolloff**
- **Skewness** (symmetry of a spectral envelope)
- **Spread** (concentration of spectrum around its centroid)

Spectral descriptors - spectral centroid

- Weighted mean of spectrum (weighted by its bin strength)
- Centre of gravity of the spectrum

$$Centroid = \frac{\sum_{n=0}^{N-1} f(n) x(n)}{\sum_{n=0}^{N-1} x(n)}$$

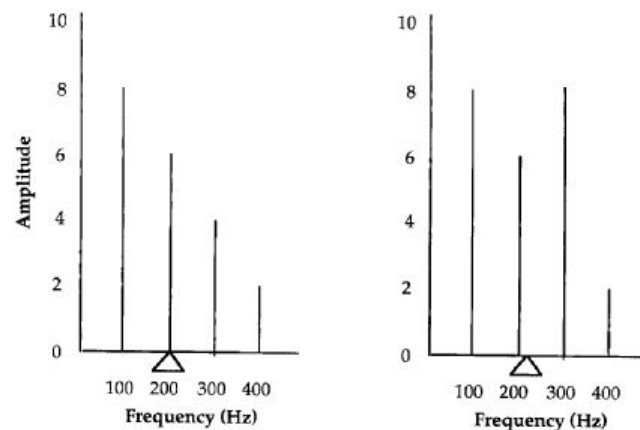
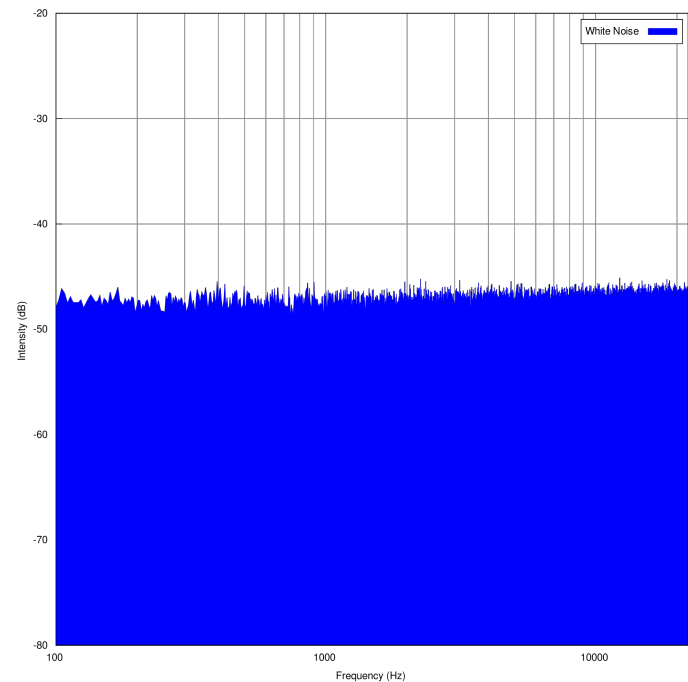


Figure 40. Two spectra with different centroids, the first at 200 Hz., the second at 216.7 Hz.

Spectral descriptors - spectral flatness

- Geometric Mean / Arithmetic mean
- It is a measure of the noisiness of a signal.

$$\text{Flatness} = \frac{\sqrt[N]{\prod_{n=0}^{N-1} x(n)}}{\frac{\sum_{n=0}^{N-1} x(n)}{N}} = \frac{\exp\left(\frac{1}{N} \sum_{n=0}^{N-1} \ln x(n)\right)}{\frac{1}{N} \sum_{n=0}^{N-1} x(n)}$$



Spectral descriptors - MFCC's

- Mel Frequency Cepstrum Coefficients:
 - A compact, perceptually motivated representation of the spectrogram
 - One of the de facto ways to describe and compare “timbre”.
- Steps
 - Take the [Fourier transform](#) of (a windowed excerpt of) a signal.
 - Map the powers of the spectrum obtained above onto the [mel scale](#), using [triangular overlapping windows](#).
 - Take the [logs](#) of the powers at each of the mel frequencies.
 - Take the [discrete cosine transform](#) of the list of mel log powers, as if it were a signal.

Spectral descriptors - MFCC's

- 40 coefficients typically computed. We usually hold onto the first 13 of them.
- Originally used in speech recognition.
- Surprising robust descriptor for musical purposes.
- Applications:
 - Genre Detection (Tzanetakis et al., 2001)
 - Instrument Identification (Herrera et al., 2003)
 - MFCCs on recorded audio captures the timbre of the recording or so-called “album effect”.

Spectral descriptors - MFCC's

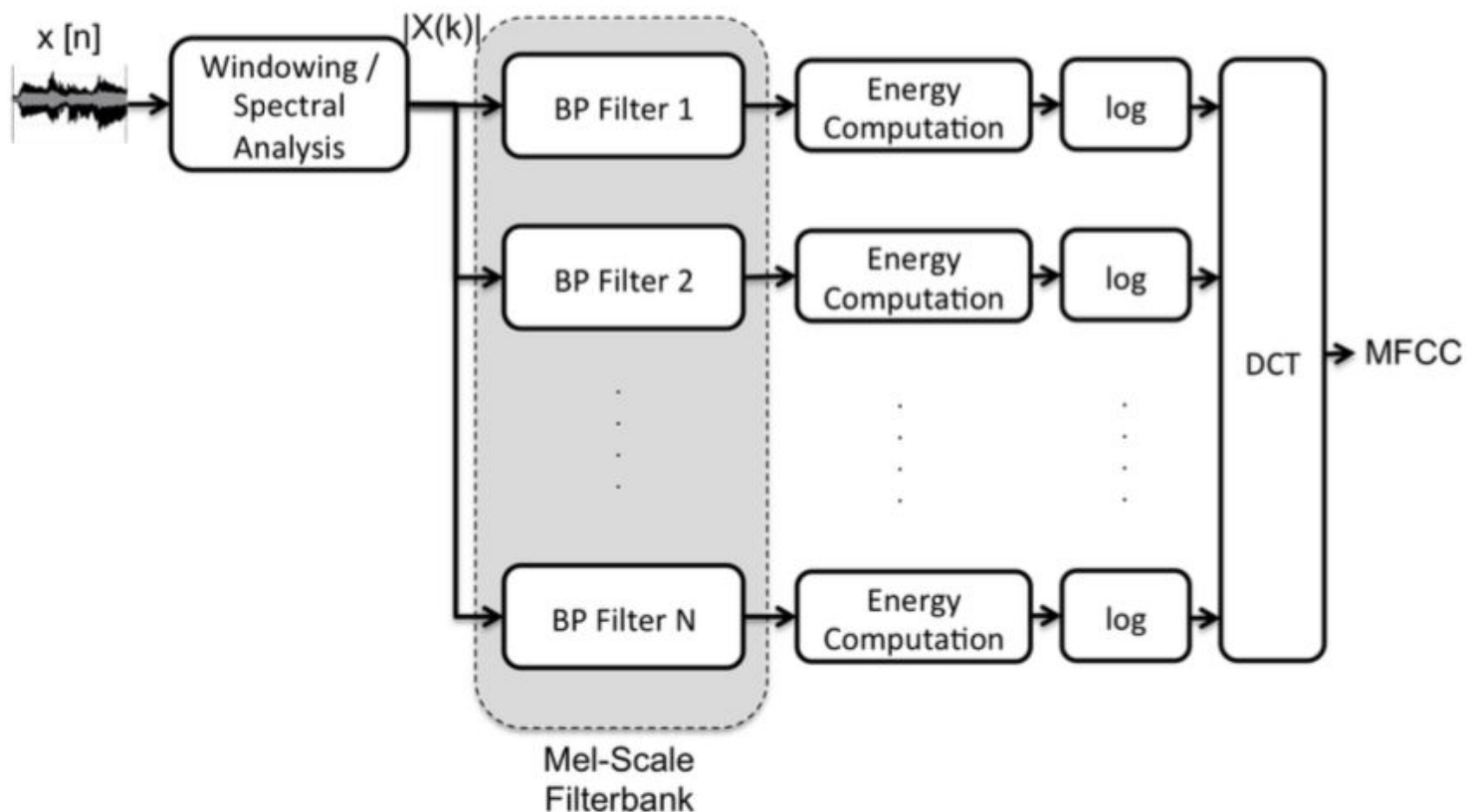


Figure 2.6: Block diagram for the computation of MFCCs

Derived and higher-level descriptors

- Melody Extraction

- f_0 (fundamental frequency)

- Tempo Descriptors

- Onset Detection
- BPM Detection
- Beats Detection

- Harmony Analysis

- Chord Detection
- Key Estimation

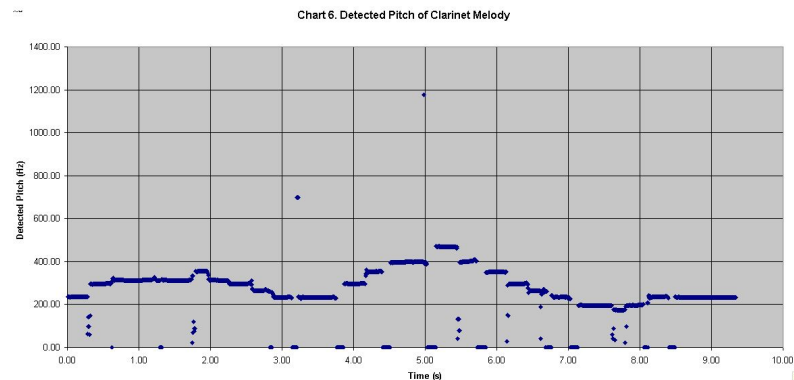
- Structural Detection

- Verse/Chorus
- ABBA...

Melody Extraction - fundamental freq. detection

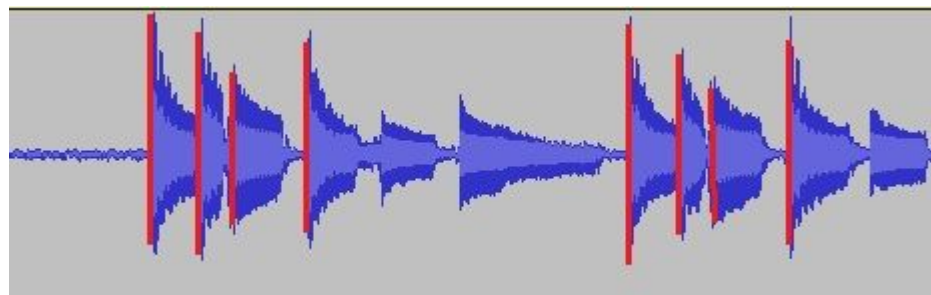
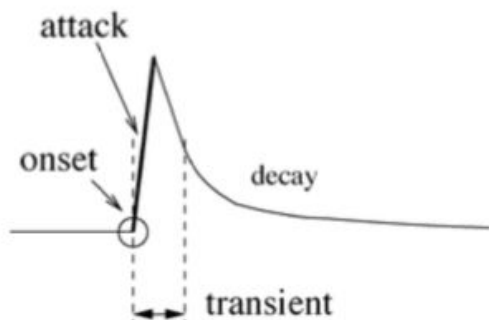
- Techniques typically involve some variation of autocorrelation
- Compare the signal against itself at ever increasing time lags
- The YIN algorithm (de Cheveigne 2001) is a variation of autocorrelation with some error suppressing

$$R_{yy}(l) = \sum_{n \in \mathbb{Z}} y(n) \bar{y}(n - l).$$



Tempo descriptors - onset detection

- Many approaches exist for detecting start of notes in signals (Bello et al. 2005)
- Typical stages include:
 - Reduce signal to onset detection function (e.g. estimating changes in spectrum from frame to frame)
 - Perform peak peaking to find local maxima corresponding to onsets

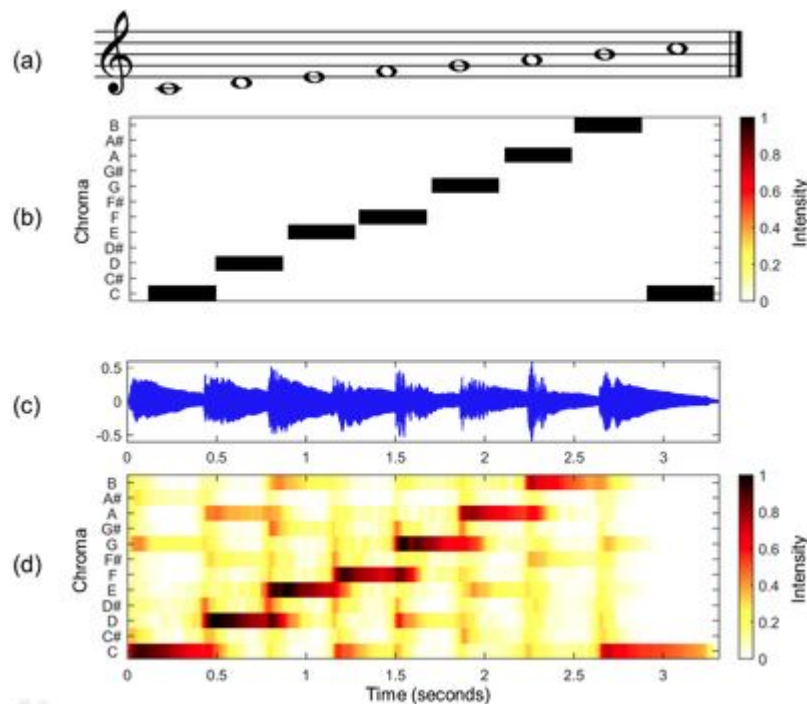


Tempo descriptors - Beats and tempo

- Beats != Onsets
- **Onsets** are where “notes” start
- **Beats** are “where you tap your foot”
- **BPM** = Beats Per Minute or what we regard as “tempo”
- Techniques include using autocorrelation on the onsets or beats to find peaks where temporal patterns emerge.

Chroma descriptors

- Extract intensities in each of the N pitch classes
- Typically 12, but there is no theoretical limitation to N-equal divisions of the octave



Aggregation

- If our analysis is frame-based, we generally want to get an “overall” impression rather than trying to manage frame by frame
- Central Tendencies
 - Mean and Variance are standard
 - But not always appropriate for every descriptor
 - For some parameters, it works better to consider mode or median

Normalisation

- Feature vectors will mostly likely comprise a combination of discrete features
 - e.g. 13 MFCCs + 1 Spectral Centroid + 1 Spectral Flatness
- Some dimensions dimension may have a large range and overpower the others
 - MinMax Normalisation
 - Standardisation: remove mean (make zero), scale by STD

$$X_{i, 0 \text{ to } 1} = \frac{X_i - X_{\text{Min}}}{X_{\text{Max}} - X_{\text{Min}}}$$

$$x_{\text{new}} = \frac{x - \mu}{\sigma}$$

Classification

- Features are extracted, aggregated and normalised.
- Now is the fun part, using your data to perform a task.
- One of these tasks may include classification - to which category does an instance of analysed data belong to.
- MIR classification examples:
 - Instrument identification
 - Genre identification
 - Chord recognition

Classification Steps

1. Gather dataset for ***training***
2. Label dataset to determine categories
3. Extract features for training set
4. Preprocess data (aggregation & normalisation).
5. Train your classifier
6. Find **test** instance or dataset
7. Extract features for test instance
8. Preprocess
9. Use classifier to identify which category the instance belongs

Mid-Level Analysis Objects in Pd

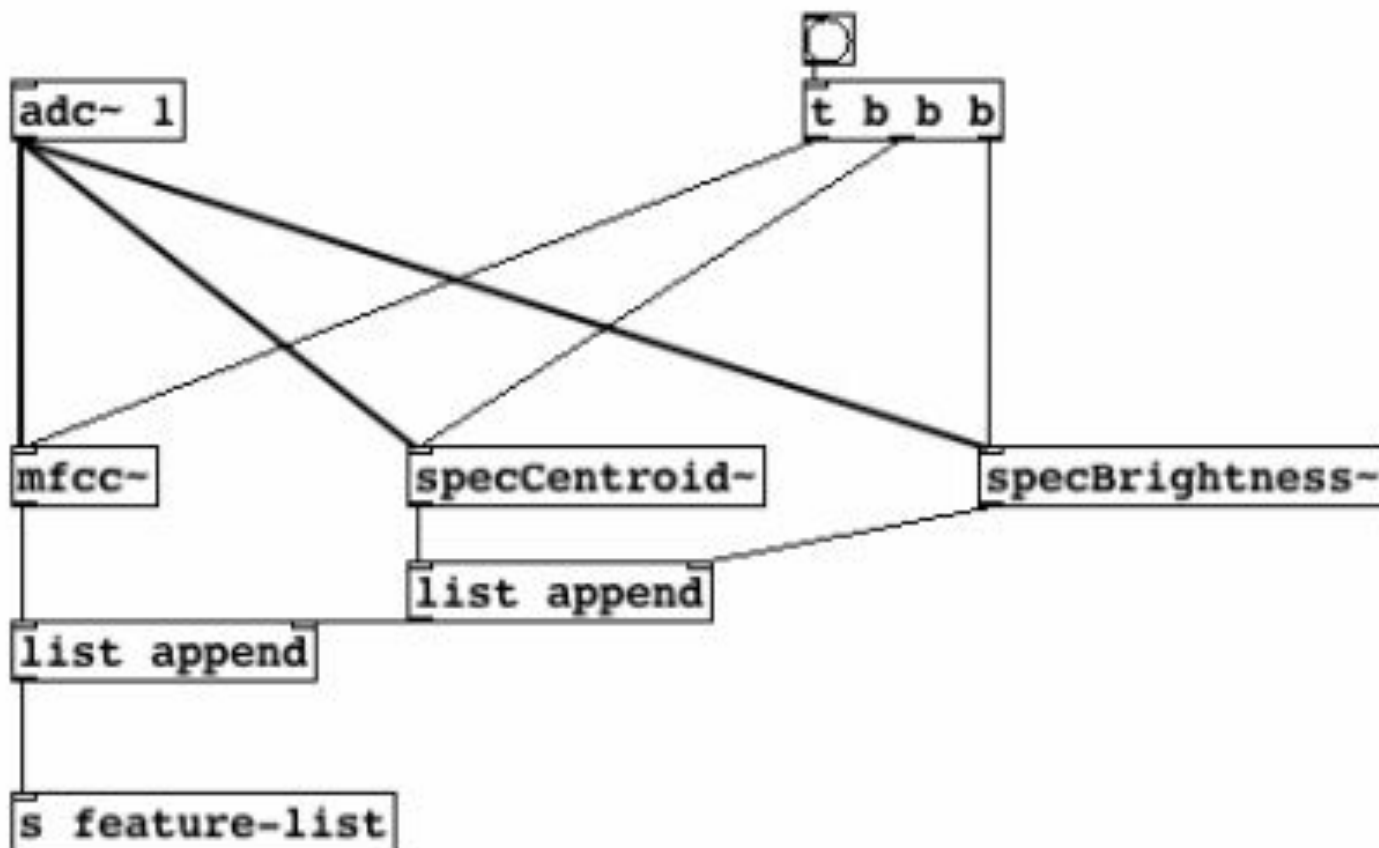
- **[env~]** RMS Amplitude follower. The RMS value of a set of values is the square root of the arithmetic mean of the squares of the original values.
- **[fiddle~]** Monophonic pitch follower.
- **[sigmund~]** Sinusoidal component analysis.
- **[bonk~]** Onset detector, mainly for percussive sounds.

Audio Descriptors and Pd: timbreIDLib

- Host of features including MFCCs, BFCCs (bark bands)
- [timbreID] object for classification and clustering
- Easy to use!

Brent, W. (2010). A timbre analysis and classification toolkit for Pure Data. In *International Computer Music Conference* (pp. 2–7). Retrieved from <http://williambrent.conflations.com/papers/timbreID.pdf>

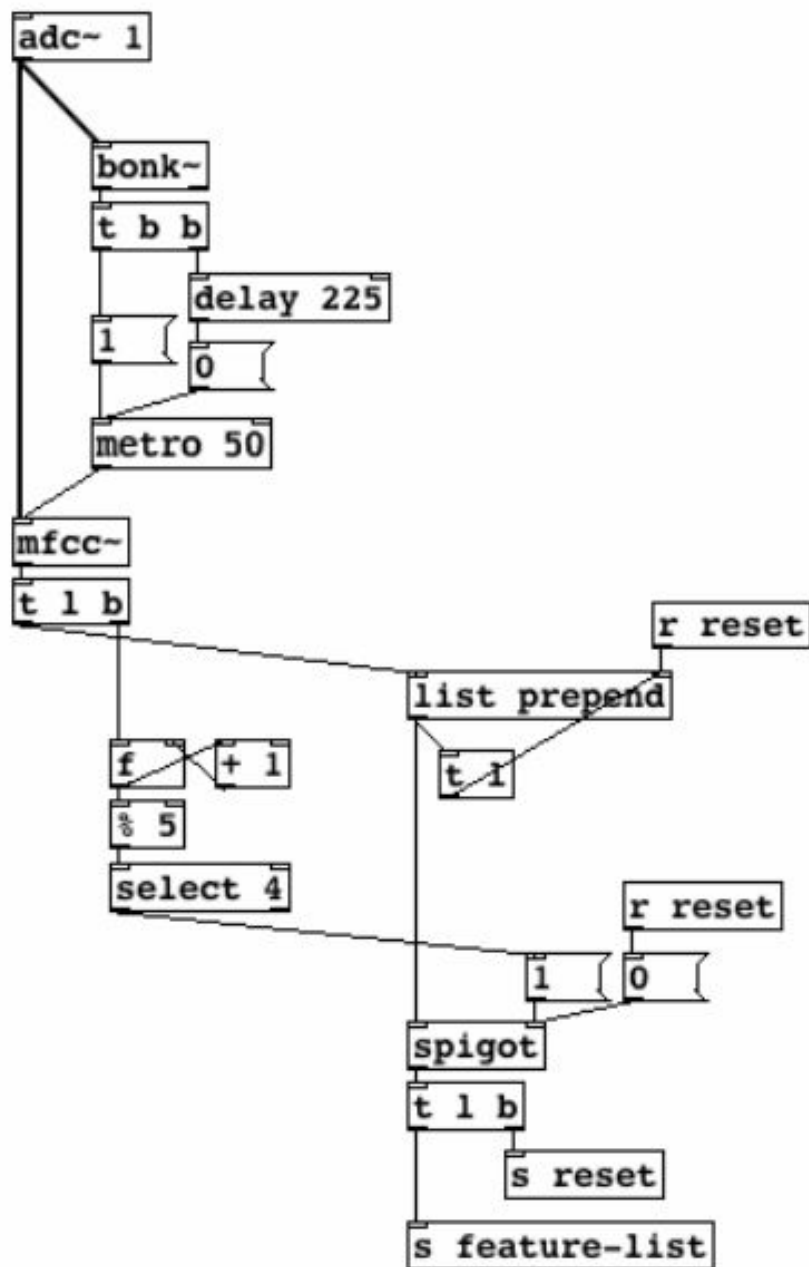
timbreIDLib - Feature aggregation



timbreIDLib - Architecture

- Each object is a feature
- Performs its own windowing, overlapping and FFT
- Send a bang message to compute features
- Use a metro object to compute features over time
 - $44100/1024 = 0.043$ ms
 - 50 ms or so is a reasonable rate.

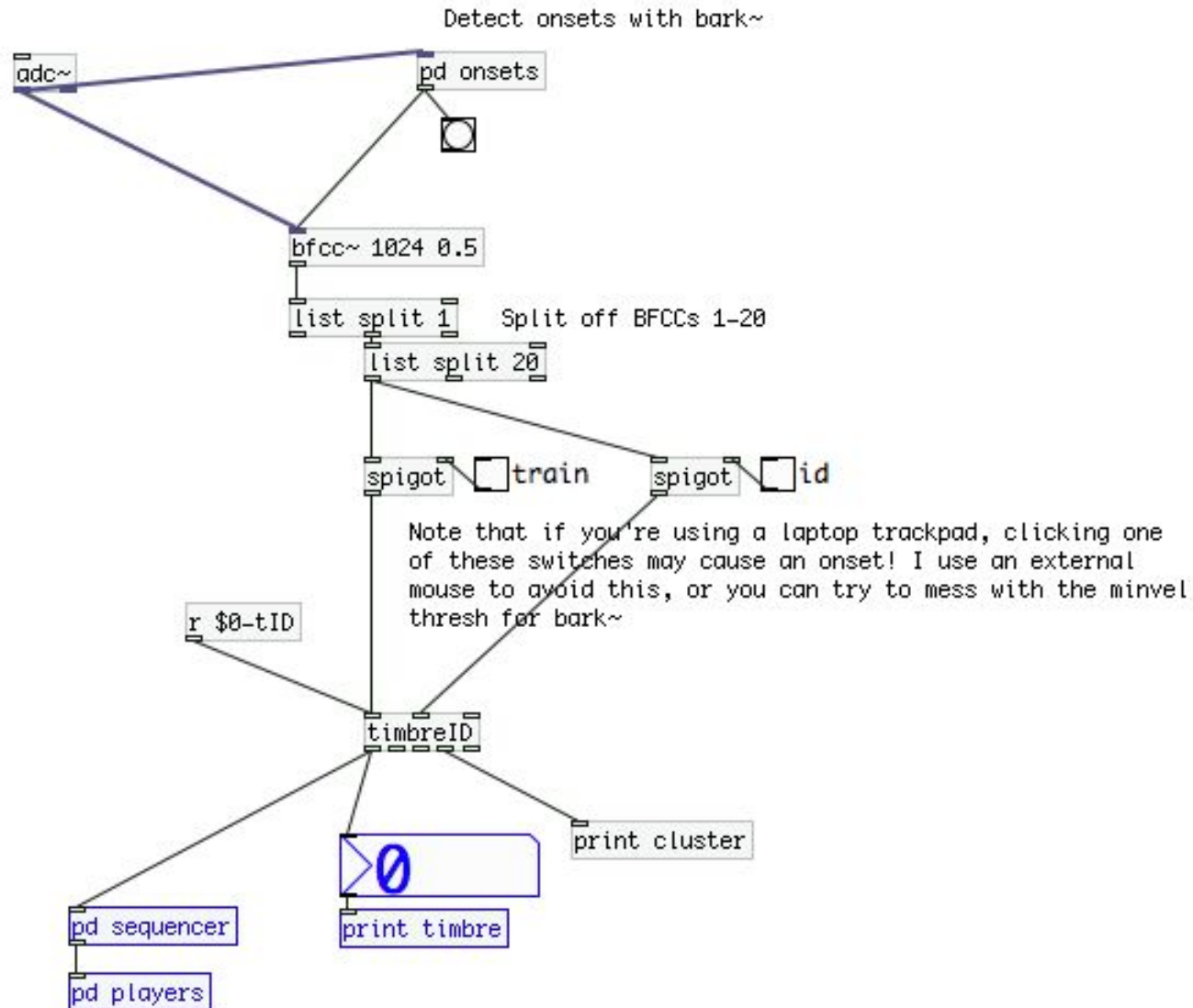
timbreIDLib



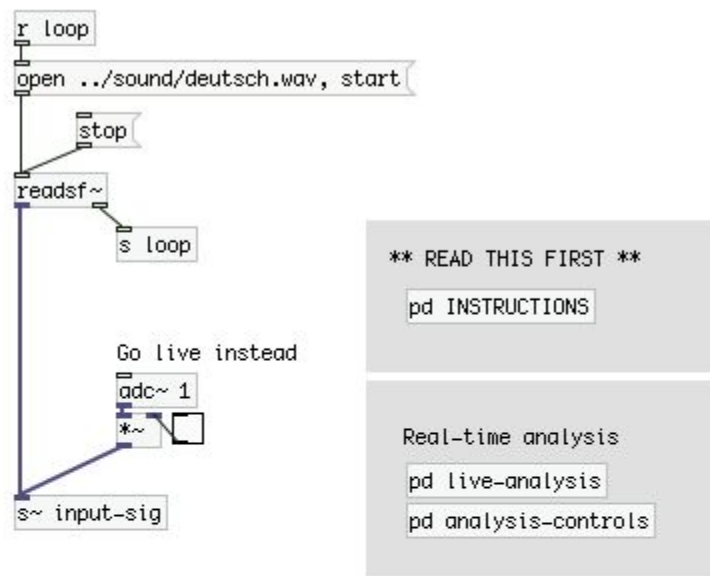
timbreIDLib - Classification

- timbreID can perform basic classification of sounds based on feature analysis (kNN and kMeans).
- Procedure:
 - a. Use an onset detector to generate bangs that can trigger analysis
 - b. Connect the output to the first inlet of [timbreID] for training
 - c. When training is complete, cluster to create the desired categories
 - d. Repeat a & b for testing but this time connecting to the second inlet of [timbreID]

timbreIDLib - Classification



timbreIDLib - Concatenative Synthesis



**** READ THIS FIRST ****

pd INSTRUCTIONS

Real-time analysis

pd live-analysis

pd analysis-controls

Mixing controls

pd input-monitor

pd granular-out

pd master-out

Master controls

☐ Load-default-sound

pd load-source-sound

☐ Load-other-sound

☐ Analyze

pd analyze-source-sound

☒ Normalize

☒ Analyze-input

☐ Env-following

pd env-follower

pd MORE-INFO

timbreID - William Brent 2011

pd init